Three ways to share one GPU, and what each one isolates

For HPC and ML platform teams running shared NVIDIA GPUs

MIG gives each job its own slice of the GPU, memory paths included

01 **Up to seven instances**

A100, H100, H200 and B200 split into up to seven GPU instances, each with separate, isolated paths through memory.

02 **Fault isolation**

NVIDIA says MIG gives a defined quality of service with fault isolation. It needs an Ampere GPU or newer.

Source: NVIDIA MIG User Guide, pulled 2026-10-31

MPS runs processes side by side, but a fatal fault forces every client on that GPU to exit

01 **Separate address spaces**

MPS clients get fully isolated GPU address spaces, and by default up to 60 client CUDA contexts per device.

02 **Limited error containment**

A fatal fault is reported to every client on the affected GPU, and those clients must exit.

Source: NVIDIA MPS documentation, pulled 2026-10-31

Time-slicing works on GPUs without MIG but isolates neither memory nor faults

01 **Works on older cards**

It gives shared access on GPU generations that do not support MIG.

02 **No isolation**

NVIDIA's GPU Operator docs state there is no memory or fault isolation between replicas.

03 **Combine with MIG**

Time-slicing can also share a single MIG instance.

Source: NVIDIA GPU Operator documentation, pulled 2026-10-31

In Slurm, MIG instances act like GPUs, but you partition them first

01 **Pre-partitioned**

Slurm 21.08+ treats MIG instances as GPUs with cgroup isolation, but needs them partitioned first.

02 **No usage accounting**

Slurm reports no gpumem or gpuutil for MIG devices, because NVML does not provide them.

03 **Shards**

Shard GRES shares a GPU without fencing processes, so it suits homogeneous work.

Source: slurm.schedmd.com/gres.html, version 26.05

What we would tell a team starting today

> **Pick the sharing mode by the isolation your jobs need, then check what your scheduler can account for.**

Which GPU sharing mode does your cluster run, and why?

Scoping is short, fixed and ends in a written plan.

contact@hopline.co